

- (iii) On Fig. 1.2, sketch the variation of v_H with time t between $t = 0$ and $t = 3.2$ s. Label your line H.

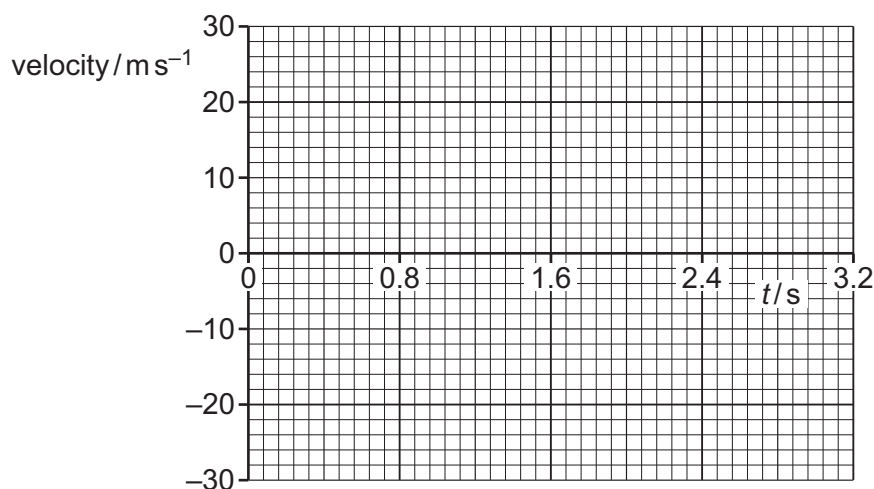


Fig. 1.2

[1]

- (iv) On Fig. 1.2, sketch the variation of v_V with time t between $t = 0$ and $t = 3.2$ s. Assume that velocity in the upward direction is positive. Label your line V. [3]
- (b) The total change in momentum of the ball between leaving the ground at $t = 0$ and landing on the ground at $t = 3.2$ s is 13 kg ms^{-1} .

- (i) Define momentum.

.....
 [1]

- (ii) Calculate the force that acts on the ball while it is in the air.

force = N [2]

- (iii) Determine the mass of the ball.

mass = kg [1]

[Total: 11]

[Turn over]

